ST MARY'S DSG, KLOOF

GRADE: 10

DATE: 30 AUGUST 2023

PHYSICAL SCIENCE OPEN BOOK TEST
KINEMATICS

TIME: 43 MINS

TOTAL: 51

ASSESSOR: K. Bolton

MODERATORS: G. Main and T. Kirsten

NAME: Meme

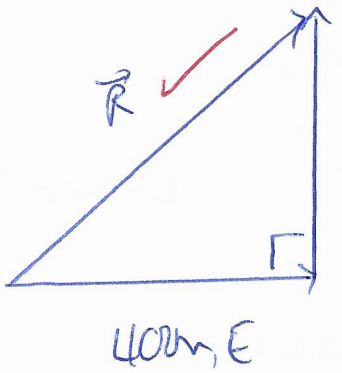
TEACHER:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 6. Please check that your question paper is complete. A data sheet is provided.
2. Answer the questions on your question paper.
3. Work to 2 decimal places for final answers.

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1. Peter walks to school 400 m **eastwards** down the street, and then he turns north and walks another 300 m due **north**.
   a. Draw a head to tail diagram to illustrate Peter's resultant displacement. (2)



head to tail

   b. Calculate the magnitude of Peter's resultant displacement. (2)

$\vec{R} = \sqrt{400^2 + 300^2}$

$= 500m$

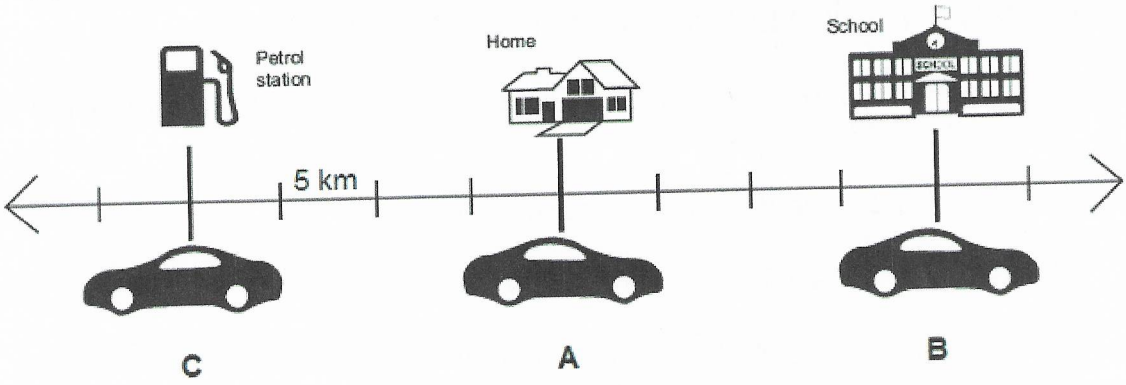
c. How far does he walk to school? (2)

700m ✓

d. When he returns home in the afternoon, what is his resultant displacement? (2)

0m ✓

2. Ted drives his car along the road, starting at point **A** and driving to point **B**, in 30 minutes. He then stops and drives from point **B** to point **C** in 45 minutes. Take the positive direction as moving to the **right.**
Scale: Each interval represents 5 km.



Calculate, in $km \cdot h^{-1}$:

a. Ted's average speed from A to B. (3)

$5 \times 4 = 20km$ ✓

$speed = \frac{20}{\frac{30}{60}}$ ✓ $= 40 km.h^{-1}$ ✓

b. Ted's average velocity from A to B. (2)

$V = 40 km h^{-1}$ ✓ right ✓

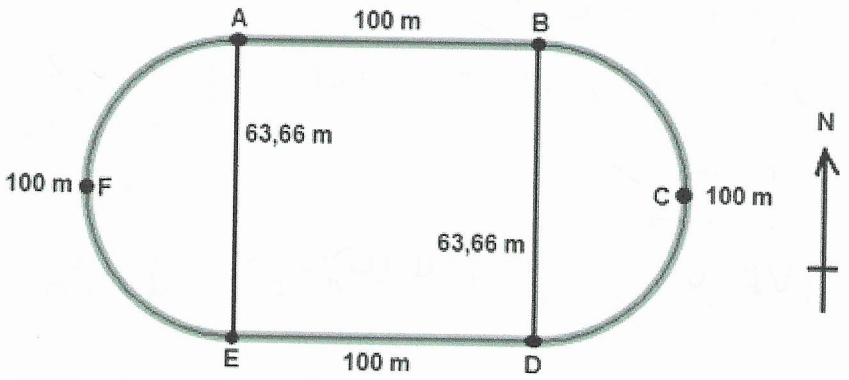
c. Ted's average speed from B to C. (3)

$$\text{Speed} = \frac{8 \times 5}{\frac{45}{60}} = 53{,}33 \text{ km.h}^{-1}$$

d. Ted's average velocity from B to C. (3)

$$V = \frac{\Delta x}{t} = \frac{8 \times 5}{\frac{45}{60}} = 53{,}33 \text{ km.h}^{-1} \text{ left}$$

3. An athlete runs **clockwise** from A to B to C to D to E on a 400 m oval athletic track at a constant speed of 5 $m \cdot s^{-1}$.



a. How long does she take to run from A to E along the track? (2)

$$t = \frac{x}{v} = \frac{300}{5} = 60 \text{ s}$$

b. Describe her displacement relative to point A when she reaches E. (2)

63,66 m, south (180°)

c. Calculate the magnitude of her average velocity when she reaches point E. (2)

$$V = \frac{63{,}66}{60} = 1{,}061 \text{ m.s}^{-1}, S\ (180°)$$

d. Determine her instantaneous velocity at point C. (2)

$5 m \cdot s^{-1}$, S (180°)

4. A tractor, travelling **east**, accelerates uniformly from rest to 4 $m \cdot s^{-1}$ in 20 s. It then travels at a steady speed for 440 m, and finally comes to rest 10 s later, having travelled a total of 500 m.

a. Calculate the acceleration of the tractor during the first 20 s. (3)

$v_f = v_i + at$

$4 = 0 + (a)(20)$

$a = 0,2 \ m \cdot s^{-2}$, E

b. Calculate the magnitude of the tractor's displacement in the first 20 s. (3)

$\Delta x = v_i t + \frac{1}{2} a t^2$

$= 0 + \frac{1}{2}(0,2)(20)^2$

$= 40 m$

OR $v_f^2 = v_i^2 + 2a\Delta x$

$(4)^2 = (0) + 2(0,2)(x)$

$x = 40$

OR $\Delta x = \left(\frac{v_f + v_i}{2}\right)t = \left(\frac{4+0}{2}\right)(20) = 40 m$

c. Calculate the magnitude of the tractor's average velocity for the entire period of its motion. (3)

$v = \frac{x}{t}$

$4 = \frac{440}{t}$

$t = 110 s$

$\vec{v} = \frac{500}{20 + 10 + 110}$

$= 3,57 \ m \cdot s^{-1}$

d. Calculate the acceleration of the tractor during the last 10 s. (3)

$v_f = v_i + at$

$0 = 4 + (a)(10)$

$a = -0,4 \ m \cdot s^{-2}$

$= 0,4 \ m \cdot s^{-2}$ W

OR

20 s 40 m. } 500 − 480 = 20 m left.

440

$\Delta x = v_i t + \frac{1}{2} a t^2$

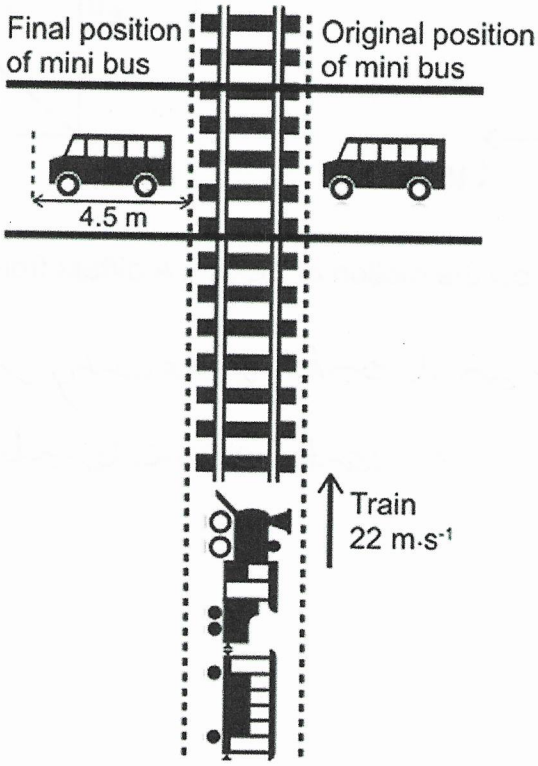
$20 = 4(10) + \frac{1}{2} a (10)^2$

$20 = 50a$

$a = -0,4 \ m \cdot s^{-2}$

$= 0,4 \ m \cdot s^{-2}$ W.

5. A train travels at a constant velocity of 22 $m \cdot s^{-1}$ along a straight railway track towards a level crossing. A mini bus, which is 4,5 m long, is stationary at the level crossing. The mini bus driver decides to cross the track. The mini bus is capable of accelerating at 2 $m \cdot s^{-2}$ and the width of the track is 2,5 m.



a. Calculate the shortest time it takes the mini bus to cross the railway track. (4)

$\Delta x = v_i t + \frac{1}{2} a t^2$ ✓

$4,5 + 2,5 = 0 + \frac{1}{2}(2)t^2$ ✓ ✓

$t = 2,65\,s$ ✓

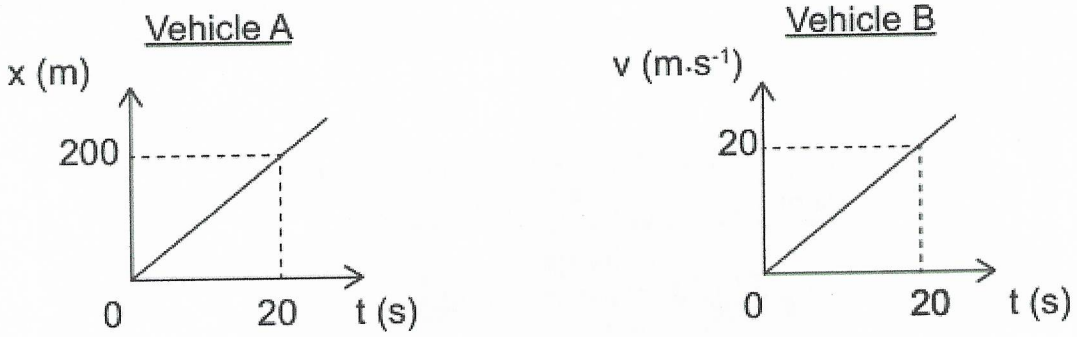
b. How far away must the train be if the mini bus is to cross the track in front of the train without colliding with it? Assume the train maintains a constant velocity. (3)

$x = v \times t$

$= 22 \times 2,65$ ✓ ✓

$= 58,3\,m$ ✓

6. The two graphs, shown below, are plotted for two different vehicles (Vehicle A and Vehicle B). Study these two graphs and answer the questions which follow.



a. Describe how the motion of Vehicle A differs from that of Vehicle B. (2)

A = constant velocity ✓

B = constant acceleration. ✓

b. How do the displacements of the two vehicles differ after 20 seconds? Which vehicle has travelled further? Justify your answer with reference to the graphs. (3)

A: $x = 200m$ ✓

B: $x = 20 \times 20$

$= 200m$ ✓

∴ equal ✓